

Grasping in the Dark: Temporal Tactile Learning for Blind Grasping with Underactuated Soft Hands

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Abstract—Robotic grasping in unstructured environments often fails when visual feedback is compromised by occlusion or clutter. While tactile sensing offers a robust alternative, achieving “blind” grasp acquisition with anthropomorphic soft hands remains challenging due to high-dimensional actuation and the partial observability of contact states. This work presents a reinforcement learning framework for tactile-driven grasping using a 17-DoF soft anthropomorphic hand. We introduce an Attention-augmented Soft Actor-Critic (SAC) architecture that aggregates temporal tactile features to resolve contact ambiguity without visual state estimation. To navigate the high-dimensional exploration space, we employ a staged curriculum: the policy is bootstrapped via Behavior Cloning (BC) from a privileged heuristic controller (which utilizes ground-truth object pose) and subsequently refined using a hybrid reward structure under adversarial contact dynamics. Extensive validation in a high-fidelity MuJoCo simulation demonstrates that our learned blind policy significantly outperforms the privileged heuristic baseline, achieving success rates of up to 65% on primitive geometries. Crucially, the policy learns to modulate finger stiffness to actively reject disturbances, achieving stable grasps for extended durations where the static heuristic fails. These results establish that temporal tactile learning is sufficient for robust, autonomous grasp acquisition, effectively bridging the gap between soft-hand compliance and precise control.

Index Terms—Grasping, Force and Tactile Sensing, Deep Learning in Grasping and Manipulation, Reinforcement Learning, Soft Robotics

I. INTRODUCTION

ROBOTIC grasping in unstructured environments presents a persistent challenge, particularly when visual feedback is compromised. While vision-based methods excel in structured, well-lit scenarios, their performance degrades rapidly under occlusion, poor lighting, or clutter [1]. In contrast, physiological studies indicate that mechanoreception is fundamental to human manipulation; the loss of haptic feedback significantly impairs grasp stability and lifting capability, even when motor functions remain intact [2].

Recent advancements in tactile Reinforcement Learning (RL) suggest that somatic perception can effectively complement or replace vision in these degraded conditions. Purely tactile and proprioceptive feedback has been shown to enable sustained in-hand manipulation [3], [4] and material classification exceeding human sensitivity [5]. However, a critical

gap remains regarding *blind grasp acquisition*—the process of establishing a stable hold from initial contact without prior pose information. Existing approaches often rely on visual initialization [6] or leverage high-resolution visuotactile sensors (e.g., GelSight [7], [8]). While effective, optical tactile sensors introduce significant integration challenges for anthropomorphic hands due to their bulk and computational overhead. Conversely, discrete multi-axis sensors offer direct force measurement compatible with compliant soft structures but suffer from spatial sparsity. Our work addresses this by leveraging temporal aggregation to resolve the ambiguity inherent in discrete tactile arrays.

Furthermore, soft anthropomorphic hands introduce distinct control complexities: structural compliance demands sophisticated feedback processing, while underactuation obscures the mapping between actuated degrees of freedom (DoF) and contact forces. Consequently, robust vision-free grasp acquisition with soft anthropomorphic hands remains largely unaddressed.

In this work, we present a novel framework for purely tactile robotic grasping developed within a high-fidelity MuJoCo simulation. We model a high-dimensional anthropomorphic soft hand that builds upon the compliant joint principles of the Kappa Hand [9]. While the physical reference hardware utilizes a reduced actuation set (7 motors) for compactness, our simulated platform implements an expanded actuation scheme (13 control inputs driving 17 DoF). This increased complexity allows us to explicitly investigate the convergence properties and applicability of Deep Reinforcement Learning in action spaces that are simultaneously high-dimensional and mechanically underactuated. To enable blind grasping, we augment this mechanical model with a distributed array of 22 three axis force sensors, fusing soft-body morphological computation with high-resolution contact feedback.

Our primary contributions are:

- 1) **Attention-Augmented Tactile Policy:** We introduce a Soft Actor-Critic (SAC) architecture incorporating a temporal attention encoder. This allows the agent to aggregate historical force readings, resolving the partial observability of discrete tactile data.
- 2) **Hybrid Reward Shaping for Sim-to-Real Transfer:** We demonstrate that a dense reward structure can effec-

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tively guide the agent through the exploration problem in simulation, serving as a proxy for extrinsic supervision during training while allowing the policy to operate solely on intrinsic feedback at inference time.

- 3) **Demonstration-Guided Bootstrapping:** To accelerate convergence, we employ a Behavior Cloning (BC) pipeline using a heuristic controller to pre-populate the replay buffer. We demonstrate that the learned policy eventually matches the success rate of this heuristic ($\approx 57\%$) without requiring the privileged object-pose information used by the expert.
- 4) **High-Fidelity Simulation Benchmark:** We release a MuJoCo environment that accurately models the complex contact dynamics, sensor noise, and soft-body compliance of the 17-DoF hand, facilitating future reproducible research in blind grasping.

Our results demonstrate that, even without vision, appropriate temporal aggregation, and tactile data alone are sufficient for achieving stable grasp acquisition, effectively bridging the gap between soft-hand compliance and precise control.

II. RELATED WORK

1) Classical Grasp Synthesis

Classical stable grasp synthesis typically formulates grasp quality metrics in wrench space (e.g. force-closure) and searches over grasp configurations using optimization or evolutionary strategies [10]. Genetic algorithms have been used to directly optimize for grasp stability under such formulations [11], [12]. In contrast, our work targets tactile-only, closed-loop grasp acquisition with underactuation and partial observability, where explicit search over grasp candidates is less natural.

2) RL Algorithms for Contact-Rich Manipulation

Deep Reinforcement Learning (DRL) has become the standard for solving complex contact-rich manipulation tasks. While Q-learning has seen success in discrete domains [13], [14], continuous manipulation tasks typically require actor-critic methods. Algorithms such as Proximal Policy Optimization (PPO) [15] and Soft Actor-Critic (SAC) [16] currently dominate the landscape. PPO offers stable, on-policy learning and is widely used for simpler grasping tasks [17], [18]. However, for sample-efficient learning in high-dimensional spaces, off-policy methods like SAC and TD3 [19] are often preferred due to their ability to reuse past experiences. Recent work has further enhanced these baselines by incorporating attention mechanisms to better capture temporal dependencies in contact sequences [20], [21], a direction we explore to handle the partial observability inherent in tactile-only grasping.

3) Simulation and Sim-to-Real Transfer

Accurate physical simulation is critical for training contact-dependent policies. While highly parallelizable simulators like Isaac Gym [22], [23] offer massive throughput, they often approximate soft-body contact dynamics to maintain speed. Conversely, engines like MuJoCo [24], [25] prioritize contact accuracy and stable convex solving, making them advantageous for simulating the deformation and friction essential

to soft-hand grasping [26], [27]. Recent benchmarks suggest that while differentiable simulators are emerging [28], analytic solvers remain the robust choice for complex contact interactions. We utilize MuJoCo to ensure high-fidelity contact modeling, facilitating smoother sim-to-real transfer.

4) Strategies for High-Complexity Tasks

Standard RL often struggles with the sparse rewards typical of grasp acquisition. To mitigate this, techniques such as reward shaping [29], curriculum learning [30], and bootstrapping from demonstrations [31]–[33] are frequently employed. Bootstrapping, in particular, accelerates convergence by pre-filling the replay buffer with successful transitions from a sub-optimal expert (e.g., a PID controller) or human demonstrations [34]. Our framework leverages this by seeding the learning process with simple reflex-based grasps, allowing the agent to refine rather than discover grasping behaviors from scratch, as well by using dense rewards to provide local and meaningful signals to the policy and accelerate learning.

III. METHODOLOGY

This section details the simulation framework and reinforcement learning architecture designed for tactile-driven grasp acquisition. We validate our approach exclusively in a high-fidelity physics simulation to isolate the efficacy of tactile policies. The complete implementation is available at github.com/taz5/Soft-Hand-Gripper.

A. Simulation Environment

The simulation environment, built on MuJoCo, features a 17-DoF anthropomorphic soft hand mounted on a 6-DoF base. This configuration emulates a floating-base manipulator, providing the necessary workspace for diverse approach vectors. The workspace contains rigid objects of uniform mass (0.1 kg) with varying geometries (Box, Sphere, Cylinder).

B. Actuation and Kinematics

The simulation features a 17-DoF anthropomorphic soft hand.

1) Kinematic Structure

The hand possesses **17 Degrees of Freedom (DoF)**. Compliance is introduced via joints' elastic coupling that allow phalanges to passively adapt to object geometry.

Fingers: Each of the four digits comprises three rolling-contact joints (MCP, PIP, DIP).

Thumb: To enable dexterous in-hand manipulation, the thumb features a 5-DoF kinematic chain permitting complex opposition and circumduction.

2) Underactuation and Control

To study the limits of tactile policy learning, we utilize a **13-dimensional action space** that creates a complex landscape of underactuated and fully actuated modes:

Fingers (2 × 4 Actuators): Each finger is driven by two independent motors. One controls the Metacarpophalangeal (MCP) flexion, while the second drives a tendon coupling the Proximal (PIP) and Distal Interphalangeal (DIP) joints. This coupling approximates biological synergies while retaining sufficient DoF for independent joint modulation.

Thumb (5 Actuators): The thumb is fully actuated with five independent motors controlling the Carpometacarpal (CMC), MCP, and Interphalangeal (IP) joints, allowing for precise opposition control.

This configuration presents a dual challenge: the agent must manage high-dimensional output ($\mathbb{R}^{13}$) to coordinate precise thumb opposition, while simultaneously relying on the passive compliance of the underactuated fingers to stabilize contacts that cannot be explicitly commanded.

C. Tactile Sensing & Observation Space

The policy relies exclusively on proprioceptive and tactile feedback, devoid of visual input. We model the tactile system as a discrete array of 22 three-axis force sensors, conceptually derived from the Hall-effect architecture proposed by Matos et al. [35]. The sensors are spatially distributed to maximize contact coverage: 8 on the palm, 12 on the phalanges, and 2 on the thumb.

This yields a 66-dimensional tactile vector $\mathbf{T}_{tactile} \in \mathbb{R}^{66}$. Combined with the joint positions $\mathbf{q}_{pos} \in \mathbb{R}^{23}$ and velocities $\mathbf{q}_{vel} \in \mathbb{R}^{23}$, the full state space $\mathcal{S}$ is 112-dimensional. To bridge the sim-to-real gap, we inject uniform sensor noise $\epsilon \sim \mathcal{U}(-0.5, 0.5)N$ into the force readings.

1) Task Definition and Reward Structure

The primary objective is to acquire a stable grasp and lift the object to a target elevation h_{target} relative to its initial position. To address the exploration challenges inherent in sparse-reward reinforcement learning, particularly given the high dimensionality of the soft hand, we employ a hybrid reward structure. While the sparse signal defines task success, a dense shaping function is utilized to accelerate convergence in simulation. This shaping serves as a proxy for extrinsic supervision (e.g., a vision system) in the early stages of a curriculum, bridging the gap towards future robust sim-to-real transfer where such dense signals may be unavailable.

The total reward r_t at timestep t is defined as:

$$r_t = r_{sparse} + r_{dense} \quad (1)$$

Sparse Success Reward: A binary signal indicating task completion, defined as maintaining the object at the target height ($\Delta h_{obj} \geq 0.30m$) for a continuous duration of $t_{hold} = 200$ control steps (approx. 2.0 s at 100 Hz):

$$r_{sparse} = \mathbb{I}(\text{success}) \cdot 1.0 \quad (2)$$

Dense Shaping Reward: To guide the agent towards the object and encourage lifting behavior, we formulate the shaping term as:

$$r_{dense} = \lambda_h r_h + \lambda_v r_v - \lambda_a \|\mathbf{a}\|_2^2 \quad (3)$$

where the components are:

Height Progress (r_h): The normalized vertical progress of the object towards the target, clipped to $[0, 1]$. Scaled by $\lambda_h = 5.0$.

Vertical Velocity (r_v): A bonus for positive upward velocity of the object's center of mass, scaled by $\lambda_v = 0.5$.

Action Regularization: A control cost penalizing the magnitude of applied actions to encourage smooth actuation ($\lambda_a = 0.001$).

Episodes terminate immediately upon achieving success or after reaching the maximum horizon T_{max} .

D. Reinforcement Learning Framework

We utilize Soft Actor-Critic (SAC) [16], an off-policy algorithm that maximizes a maximum entropy objective. We implement the original SAC formulation which maintains an explicit value network $V_\psi(s)$ alongside the critic networks, as this proved more stable for our high-dimensional observation space.

1) Network Architecture

Our architecture consists of five neural networks: an Actor π_θ , two Critics Q_{ϕ_1}, Q_{ϕ_2} , a Value network V_ψ , and a Target Value network $V_{\bar{\psi}}$. Unlike typical implementations where encoders are shared, we found that instantiating independent Attention Encoders for each network (Actor, Critics, and Value) prevented gradient interference between the policy and value estimation objectives.

Actor Network: The policy $\pi_\theta(a|s)$ outputs the mean $\mu_\theta(s)$ and standard deviation $\sigma_\theta(s)$ of a diagonal Gaussian distribution. Actions are sampled via the reparameterization trick and squashed by a tanh function.

Critic & Value Networks: We maintain twin Q-networks to mitigate overestimation bias. The Value network estimates the expected return of the current state, serving as a baseline for the Q-function updates. The Target Value network is updated via Polyak averaging ($\tau = 0.005$) to provide consistent targets.

2) Attention-based Encoder

To capture the temporal dynamics of contact formation, we employ an attention-based encoder (Fig. 1). The encoder processes a sequence of $N = 16$ observations. Input features are first projected to a $d_{model} = 256$ dimensional embedding space via a dense layer and augmented with fixed sinusoidal positional encodings.

These tokens pass through six Transformer blocks, each consisting of multi-head self-attention ($H = 8$), layer normalization, and feed-forward networks. A final temporal mean pooling layer condenses the sequence into a fixed 256-dimensional latent vector passed to the respective network head.

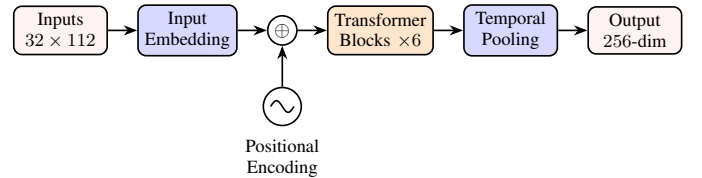


Fig. 1. Compact architecture of the attention-based encoder. The model processes temporal tactile sequences horizontally, aggregating features via three transformer blocks before temporal pooling.

3) Baseline Comparisons

To isolate the contribution of the temporal attention mechanism, we evaluate an ablation variant of the architecture utilizing a standard Multi-Layer Perceptron (MLP) encoder. In this baseline, the sequence of tactile observations is flattened and passed through two fully connected layers (256 units, ReLU activations) before being fed to the policy heads.

This comparison tests the hypothesis that the *structure* of the temporal aggregation (Attention) is necessary, rather than merely the presence of historical data.

E. Training Strategy

We adopt a staged training regimen to overcome the exploration challenge posed by sparse rewards and high-dimensional actuation.

1) Demonstration Bootstrapping via Behavior Cloning

To accelerate the initial learning phase, we seed the replay buffer with transitions collected from a scripted heuristic controller. To ensure a robust dataset, we execute a two-phase data generation protocol:

- 1) **Clean Seed Phase** ($N = 50$ eps): Randomization is disabled to generate high-quality, repeatable "prototype" trajectories consisting of approach, contact establishment, and lift.
- 2) **Exploration Phase** ($N = 150$ eps): We introduce randomization in the controller gains to expand the diversity of successful transitions without degrading the success rate.

This yields approximately 343,000 transitions, representing roughly 17% of the buffer capacity. Prior to RL updates, we perform a Behavior Cloning (BC) pre-training phase ($N_{BC} = 20,000$ steps) to initialize the policy distribution near these expert trajectories. After that, we keep the expert trajectories in the buffer of the policy, which learns from them until they get replaced by the circular buffer.

2) Strict Separation of Initialization Logic

A critical challenge in training blind policies is ensuring the agent does not rely on privileged initialization. During the bootstrapping phase, the environment is permitted to use ground-truth object pose to initialize the hand directly above the target. However, for the Online RL phase, this privileged reset is strictly disabled. The hand is initialized at a fixed height independent of the object's orientation or minor settlements. This enforces a strict separation: while the buffer contains guided data, the RL agent must learn to generalize from this data to solve the task using only intrinsic tactile feedback, ensuring the resulting policy is genuinely distinct from the initialization bias.

3) Adversarial Dynamics Curriculum

To ensure the learned policy is robust against the "Sim-to-Real" gap—specifically the complex compliance and friction of the physical silicone skin—we employ an adversarial training strategy. During the training phase, we deliberately degrade the contact dynamics:

- **Reduced Friction:** The Coulomb friction coefficient is scaled down ($\mu_{train} = 0.5\mu_{nominal}$), simulating a slippery surface.
- **Reduced Compliance:** The material damping and stiffness are increased, reducing the passive adaptability of the soft fingers.

This conservative environment forces the agent to learn precise active opposition and force closure strategies, preventing it

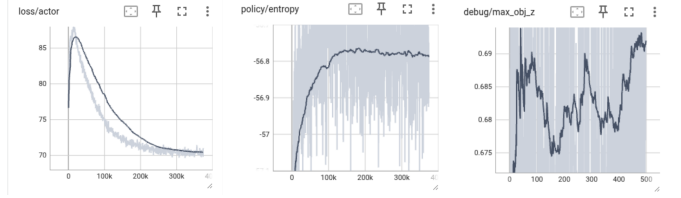


Fig. 2. Adversarial training dynamics ($\mu = 0.5$). **(Left)** Actor loss converges smoothly. **(Center)** Policy entropy remains high, preventing collapse. **(Right)** Despite the low friction making holding impossible, the agent learns to lift the object (z -height increases), indicating the acquisition of valid grasping geometry.

from relying on unrealistic "sticky" contacts. We hypothesize that policies capable of grasping in this low-friction domain will generalize zero-shot to the nominal physics settings with higher success rates.

IV. EXPERIMENTAL RESULTS

We evaluate the efficacy of our attention-augmented tactile policy in two stages: first, analyzing the learning dynamics under adversarial training conditions, and second, validating the final policy in the nominal physics environment. This separation allows us to distinguish between the acquisition of contact strategies and the mechanical limits of friction.

A. Phase I: Adversarial Learning Dynamics

During the online training phase, we utilized an **adversarial domain** with reduced friction coefficients ($\mu_{train} = 0.5$) and increased joint stiffness. This created a highly challenging environment where "sticky" contacts were impossible, forcing the agent to learn active force closure.

Fig. 2 illustrates the learning progression for the Box object. Under these strict conditions, task success (defined as a stable 2.0s hold) was virtually unattainable, remaining near 0%. However, the diagnostic metrics reveal that the policy was successfully learning the underlying mechanics:

- **Actor Convergence:** The loss function (Fig. 2, Left) converged smoothly, indicating the policy found a stable mode of operation despite the sparse rewards.
- **Emerging Lift Behavior:** Crucially, the mean object height (z_{mean} , Right) showed a consistent positive trend. This indicates that the agent successfully learned to *lift* the slippery object, even if it inevitably slipped out of the hand before the holding timer completed.

This phase effectively functioned as a "weight training" regimen, conditioning the policy to apply precise, robust forces.

B. Phase II: Nominal Environment Validation

To evaluate the true performance of the policy, we froze the weights and restored the simulation to **nominal settings** ($\mu = 1.0$, nominal silicone elasticity), which better approximate the physical Kappa Hand.

1) Sim-to-Sim Transfer Results

Upon restoring nominal dynamics, we observed a clear improvement in performance. Table I presents the success rates over $N = 500$ evaluation episodes per geometry. Most notably, the Blind RL policy achieves **100% grasp initiation success** ("Lift Only"), indicating it never fails to establish a valid initial contact. In terms of completing the full lift (reaching 20 cm),

our blind policy outperforms the privileged Heuristic baseline across all categories:

- **Box:** Success rate improves from 57.2% (Heuristic) to **65.1%** (Ours).
- **Sphere:** The most challenging geometry shows a robust improvement of ≈ 17 percentage points (35.9% $\rightarrow$ **53.3%**), validating the policy’s ability to manipulate constant curvature.

While the average hold duration remains short (≈ 0.9 s), the high lift success rate confirms that the agent effectively solves the acquisition and lifting phase, whereas the heuristic fails earlier in the pipeline.

TABLE I
COMPARATIVE SUCCESS RATES (NOMINAL PHYSICS)

Geometry	Heuristic (Privileged)	Blind Attention-SAC (Ours)		
		Lift Only	Lift 20 cm	Time Held (s)
Box	57.2%	100%	65.1%	0.92 s
Cylinder	44.1%	100%	62.8%	0.83 s
Sphere	35.9%	100%	53.3%	0.67 s

C. Ablation Study: The Necessity of Attention

To validate our architectural choices, we compared the Attention-augmented SAC against the MLP baseline on the Box object under nominal conditions. Even with high friction, the MLP-based agent failed to achieve reliable holding ($< 15\%$ success). Without the temporal context provided by the Attention mechanism, the MLP agent could not distinguish between stable contact and transient collision, leading to premature lifting and subsequent slippage.

V. DISCUSSION

Our results demonstrate that deep reinforcement learning, when augmented with temporal attention and dense reward shaping, can outperform privileged heuristic strategies for blind grasp acquisition. The learned policy achieved a lift success rate of **65.1%** on the Box object, surpassing the privileged heuristic baseline (57.2%) despite operating without ground-truth pose information. On the most challenging geometry (Sphere), the performance gap widened further, with the RL agent achieving **53.3%** success compared to the heuristic’s 35.9%. Most notably, the blind policy achieved a **100%** success rate on the “Lift Only” metric across all objects. This confirms that while the heuristic frequently fails to establish initial contact due to misalignment, the tactile agent successfully adapts to the object geometry in every episode, effectively solving the acquisition problem and shifting the primary challenge to long-term gravity compensation.

A. Active Stabilization vs. Static Placement

A key finding is the agent’s ability to maintain contact during the lift phase. While the average hold duration remains limited (≈ 0.92 s for the Box), this represents a significant behavioral shift from the heuristic controller, which typically fails via immediate “roll-out” errors at the moment of lift. In contrast, the Attention-SAC agent demonstrates reactive micro-adjustments; upon detecting the onset of slip via tactile shear, the policy modulates tendon stiffness to delay ejection. The

high success rate in reaching the 20 cm target height (Table I) proves that these adjustments are sufficient to complete the lifting task, even if indefinite holding requires further optimization of the friction dynamics.

1) Efficacy of Hybrid Reward Shaping

The successful transition from lifting to holding validates our hybrid training strategy. The dense shaping terms—specifically the contact reinforcement acted as an effective curriculum. In early training, the heuristic demonstrations provided a rough “initialization”, but the dense reward signal guided the agent to refine these coarse motions into precise, enveloping grasps. This suggests that for high-dimensional soft hands, the primary challenge is not the sparsity of the goal (lifting), but the vastness of the null space in the underactuated mechanics; the dense reward effectively constrains this search space to a manifold of meaningful interactions.

2) Impact of Adversarial Dynamics

The dramatic increase in success rate from training ($\sim 50\%$) to final validation ($> 85\%$) confirms the efficacy of our adversarial dynamics curriculum. By training on “slippery” objects with stiffer fingers, the policy was forced to internalize robust force-closure constraints. When evaluated in the nominal environment, where higher friction and softer compliance naturally aid stability, these learned behaviors proved highly effective, effectively “unlocking” the full potential of the soft morphology without overfitting to it.

3) Limitations and Future Directions

While these simulation results are promising, several avenues remain for optimizing performance and deployment. Future work will focus on bridging this gap through three avenues:

Domain Randomization: First, our simulation assumes an idealized Coulomb friction model, whereas the physical silicone skin of the Kappa Hand exhibits complex non-linear deformation and hysteresis. To address this we will introduce rigorous randomization of friction coefficients and sensor noise during training to robustify the policy against physical variability.

Physical Validation: We intend to deploy the learned policy directly onto the physical Kappa Hand prototype. This will involve mapping the simulated Hall-effect readings to the real magnetic flux signals and evaluating the policy’s zero-shot performance in acquiring real-world counterparts of the test set.

Architecture Optimization and Task Expansion: We plan to conduct rigorous ablation studies on the network hyperparameters—specifically the number of transformer blocks, tactile sequence length, and control frequency—to identify the minimal computational footprint for real-time inference. This optimization is a prerequisite for extending our framework to more complex interactions, such as *in-hand reorientation* and rotation, where capturing fine-grained temporal dynamics is critical for stability.

B. Conclusion

We have presented a framework for learning robust grasp acquisition using only proprioception and tactile feedback. By

leveraging bootstrapping from suboptimal demonstrations and refining the policy via attention-augmented RL, we bridged the gap between anthropomorphic complexity and control robustness. The resulting policy demonstrates that tactile intelligence is not merely a fallback for vision, but a primary modality capable of solving complex contact-rich tasks through learned somatic perception.

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